

Vector Database Comparison Guide

1. Overview

Vector databases are purpose-built storage systems optimized for storing, indexing, and querying high-dimensional embedding vectors. Unlike traditional relational databases, they support approximate nearest-neighbor (ANN) search, enabling semantic similarity queries at scale. The choice of vector database impacts query latency, recall accuracy, scalability, and operational complexity. This guide compares the most widely used options in 2024-2025.

2. Key Evaluation Dimensions

When evaluating vector databases, practitioners should consider: (1) ANN algorithm and recall at different latency budgets; (2) support for metadata filtering; (3) persistence and durability guarantees; (4) horizontal scalability; (5) hybrid search (dense + sparse); (6) managed cloud offering vs. self-hosted; (7) SDK ecosystem; and (8) licensing model.

3. Comparison Table

Database	ANN Algorithm	Filtering	Managed Cloud	License
ChromaDB	HNSW	Yes	No (local)	Apache 2.0
Pinecone	Proprietary	Yes	Yes	Commercial
Weaviate	HNSW	Yes	Yes (WCS)	BSD-3
Qdrant	HNSW	Yes	Yes	Apache 2.0
Milvus	IVF/HNSW	Yes	Yes (Zilliz)	Apache 2.0
pgvector	IVFFlat/HNSW	Yes (SQL)	Via Postgres	PostgreSQL
FAISS	IVF/PQ/HNSW	No (native)	No	MIT

4. ChromaDB Deep Dive

ChromaDB is the most popular choice for local and development RAG pipelines. It uses HNSW (Hierarchical Navigable Small World) graphs for ANN search, offering sub-millisecond query times for collections up to ~1 million vectors. ChromaDB supports cosine, L2, and inner product distance metrics, metadata filtering using a MongoDB-like query syntax, and persistent storage via SQLite. Its Python-first API and zero-dependency setup make it the default choice for prototyping RAG systems.

5. Scaling Considerations

For production deployments handling tens of millions of vectors or requiring high-availability, managed solutions such as Pinecone or Qdrant Cloud are recommended. For teams already

invested in PostgreSQL infrastructure, pgvector offers a compelling option as it eliminates a separate database service. Milvus is the recommended choice for on-premise deployments requiring horizontal scaling across multiple nodes.